



## Module Descriptor

**Module Code:** CN101.3

**Version:** 2/ August 2019

**Module Title:** Data Communication and Networks

**Faculty:** Computing

**Level:** 1

**Contact:**

| Module Leader        | Email                                                      | VLE                                                               |
|----------------------|------------------------------------------------------------|-------------------------------------------------------------------|
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**Delivery Pattern:**

| Credits | Contact Hours | Independent Study Hours | Total Learning Hours |
|---------|---------------|-------------------------|----------------------|
| 3       | 45            | 105                     | 150                  |

**Learning Outcomes:**

|     | Module Learning Outcome (LO)                                                                                                  | Award LO                  |
|-----|-------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| LO1 | Understanding principles of data and computer communications                                                                  | Knowledge & Understanding |
| LO2 | Understanding the layered approach for communication – OSI model                                                              | Knowledge & Understanding |
| LO3 | Demonstrate good Understanding of the Physical layer functions and protocols and designing of the physical layer of networks. | Knowledge & Understanding |
| LO4 | Demonstrate the types of Wide Area Networks and Protocols and their applications                                              | Application               |

**Module Details:**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicative Content | Data Communication Fundamentals: Communication system components, Analog vs Digital communications, Bandwidth, broadband vs. baseband. Networking Fundamentals: LAN vs WAN, Network topology, OSI model and Layer functions and protocols. Physical layer functions and mechanisms, Ethernet and IEEE 802.xx standards and protocols. Cabling options, optical fiber and other available media. Introduction to WAN and MAN technologies. |
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|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assessment Details     | Students will be evaluated on the basis of take home assignment and end-term exam. The weightage given to each of these units is listed below:<br>In-class Test-40%<br>Exam-60%                                                                                                                                                     |
| Learning strategies    | 2 hours lecture and 1-hour tutorial per week.<br>Lectures and presentations with audio-visual support, class room activities including class room exercises, individual and individual assignments which enable the students to enhance practical aspects and application of Protocols. Network configuration workshop and labs.    |
| Prospectus Information | This course aims to provide the learners with a firm understanding of how data flow from one station to another station and also provides an idea about transmission devices.                                                                                                                                                       |
| Reference Texts        | Stallings, W., Data and Computer Communications, Prentice Hall. Forouzan, B.A., Data Communications and Networking, McGraw Hill<br>Kurose, J.F & Ross, K.W., Computer Networking – A Top-Down Approach Featuring the Internet, Pearson/Addison-Wesley<br>Hallsall, F., Computer Networking and the Internet, Pearson/Addison-Wesley |
| Other Resources        | All lectures will be made available as MS-PowerPoint presentations through the Learning Management System (LMS). Network devices and tools are available in labs to practice network configurations.                                                                                                                                |

